

Group 1 Preliminary Analysis Report

Introduction:

Machine learning-based algorithms have been widely deployed in the healthcare industry to predict factors like healthcare costs and health insurance claims. While the use of these algorithms often promises efficiency, fairness, and high accuracy, optimized predictive technologies in fact lead to social harms that exacerbate the conditions of marginalized demographic groups (Wang et al., 2023). For example, a study in 2019 has shown how racial bias in an algorithm used to predict potential healthcare needs for patients lead to fewer resources being spent on African American patients compared to White patients (Obermeyer et al, 2019). Inspired by this phenomenon, our project aims to identify how biases during data collection, specifically Race, Age, and Gender, may result in the creation of biased algorithms. We will focus our examination using three different model flexibilities: Linear Regression with low flexibility, Polynomial Regression with medium flexibility, and K Nearest Neighbors Regression with high flexibility. Using these algorithms, we will answer the research question: **How do different biases during data collection and the flexibility of a model affect algorithmic biases when predicting a patient's hospital visit billing amount? From this research, we expect to find bias during data collection to negatively impact the predictive accuracy of billing amount, specifically for patients within the groups exhibiting the bias.**

Our data was collected from a Kaggle dataset titled 'Healthcare Dataset' created by Prasad Patil and contains 10,000 synthetic patient entries (i.e, randomly generated values). The data set contained variables of interest such as Blood Type: A patient's blood type, Medical Condition: what medical condition a patient is being treated for (Cancer, Obesity, Hypertension, Diabetes, and Arthritis), Insurance provider: the insurance provider a patient uses (Cigna, Medicare, BlueCross, UNH, Atena), Medication: the medication a patient was prescribed, Test Results: results from testing done during patient examination, and Admission type: the level of urgency a patient was admitted with, as well as our outcome of interest: patient billing amount.

In terms of data cleaning and preparation, given the synthetic nature of our data, we were able to skip past many of the typical steps needed for data cleaning. For instance, we did not need to handle N/A, miscellaneous, and/or repeated values/rows. While this simplified the data cleaning process, we found other cleaning tasks that needed to be completed. For example, converting all patient names in the 'Names' column to title case, and removing columns 'Room Number' and 'Doctor' that we deemed unnecessary for our analysis. The next step was creating features we hypothesized were correlated with a patient's billing amount. This included the addition of a 'Race' column, which we derived for each patient using US census data regarding the demographic makeup of the United States (e.g, 60% White, 12% African-American etc.). Additionally, a patient's Admission length (Date of Admission - Date of Discharge) seemed likely to be correlated with their billing amount, so we added an 'Admission length' column.

Finally, we wanted to analyze potential age biases that could occur during data collection so we categorized each patient via the 'Age' column in buckets ranging from 0-21 to 65+ in the 'Age Group' column.

After our initial data cleaning and preparation, we decided on our predictors of Gender, Blood Type, Medical Condition, Insurance Provider, Admission Type, Medication, Test Results, Race, Age Group, Admission Length, and Age. Given that the data was synthetic, we wanted to explore how each relevant predictor, excluding variables such as hospital and doctor, could be a statistically significant predictor of patient billing amount. One final step of data cleaning/preparation we did was after our initial Exploratory Data Analysis. We discovered that the values within our predictors needed to be converted to either categorical predictors or required dummy variables. For instance, the 'Blood Type' and 'Medical Condition' columns were multi-level string predictors that needed dummy variables to represent the different conditions and types. To solve this, we used the `get_dummies()` function from `pandas` to create columns containing the different categories and regressed using these dummy variables.

Our initial analysis of models using these predictors revealed that due to the data being synthetic, there existed no relationship between any predictors and billing amount. Thus, we had to create a correlation. To do this, we researched real-world statistics on how each feature may impact billing amount and incorporated that correlation into our data. For instance, research conducted by the World Economic Forum revealed that women on average paid 1.18x more on hospital billing compared to men (Edmond,1), so we multiplied each female's billing amount by 1.18. (Please see full description of our edits in the table below).

Methods:

There exists a vast array of approaches which could be taken to analyze our dataset, and we decided to utilize several different algorithms on many different subsets of our data in order to provide a truly comprehensive analysis of any potential bias that might be represented. Using multiple different models allowed us to compare and contrast the different algorithms, which in turn was beneficial in revealing not just the existence of bias, but also the root cause.

The first algorithm we utilized was a Linear Regression Model. This is a fantastic first choice, as it is an incredibly simple and straightforward algorithm which allows for us to create a baseline from which to compare other models. Linear regression assumes a linear relationship between predictors, and allows you to see bias in model weights directly, providing a clear understanding on how each variable influences the outcome.

The second algorithm we chose was a K-Nearest Neighbors model. This is meant to predict outcomes based on data distribution, by grouping together similar data points into clusters. KNN is particularly effective at detecting data driven bias, and by comparing it to our

linear regression model we can see how much of the model's bias comes from the data itself versus the algorithm's assumptions.

The third and final algorithm we employed was Polynomial Regression. This is a non-linear regression model that can outperform standard linear regression at predicting more complex relationships, especially when bias is not uniform across populations. Additionally, it allows us to select our own degree for the polynomial, which means we can get the best possible fit for the data.

After selecting our models, we had to format our data for training, and then evaluate each model's performance on biased subsets of our data. First, our models were trained on our original dataset, which matched real-world distributions of Age, Race, and Gender statistics. We then created biased subsets of our dataset, which were built upon unequal splits of each of these demographics. These consisted of Gender split datasets, (containing 70, 80, and 90% male participants respectively), Race split datasets (70, 80, and 90% white participants), and Age split datasets (70, 80, and 90% of participants from the 0-40 age group). To create these datasets while ensuring the original data structure was preserved, each subset was constructed by randomly sampling participants from the original dataset, until we had groupings that match our desired ratios exactly. Once all nine subsets had been created, we ran each of the three predictive algorithms on each of the subsets, measuring the R^2 value, the RMSE, and the predicted billing amount/standard deviation for each split.

By utilizing this split-dataset approach, we are able to perform controlled experiments on demographic imbalance. When we examine datasets with specific demographic ratios, we can see how the models behave as the underlying population structure changes: For example, if our model predictions change drastically with a 90/10 gender split as compared to a 50/50, we can extrapolate that gender is a sensitive characteristic which has influence over model predictions, and this raises red flags for fairness.

This approach is also useful for separating data driven bias from algorithm driven bias. Each of these models handles demographic imbalance differently, and by keeping the training data fixed and only modifying the testing population, we can determine if bias originates from the structure of the algorithm, the structure of the data, or the interaction between the two.

Results:

To answer our research question, we analyzed the effects of three distinct data collection biases (Age, Race, and Gender) on the predictive accuracy and fairness of three models with varying flexibility: **Linear Regression**, **Polynomial Regression** (Degree 2), and **KNN Regression** ($k=15$). Our analysis focused first on aggregate predictive accuracy (R-Square and MSE/RMSE) and second on algorithmic bias (group-specific error disparities).

1. Aggregate Model Performance

Across all analyses, the Polynomial Regression model was consistently the most accurate. When trained on the original, unbiased data, it achieved the highest R-Square (0.997) and the lowest Mean Squared Error (MSE) ($1.347899e+07$). The low-flexibility Linear Regression model was also a strong performer (R-Square ≈ 0.905), while the high-flexibility KNN model was the weakest (R-Square ≈ 0.659).

2. Impact of biased datasets

We then investigated how introducing bias into training data affected the performance of these models.

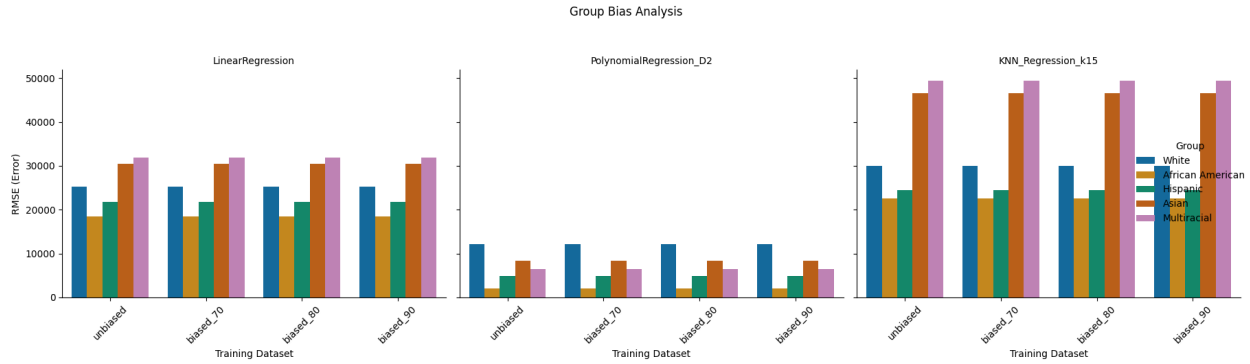
- **Age Group Bias:** We trained all three models on datasets with increasing age bias (70/30, 80/20, and 90/10 splits favoring the 0-40 age group). The overall accuracy of all three models remained remarkably stable. The **Polynomial** model's *R-Square* was ≈ 0.997 , the **Linear** model's was ≈ 0.90 , and the **KNN** model achieved **R-Square** ≈ 0.73 regardless of whether they were trained on the unbiased data or different biased splits.
- **Racial Bias:** A similar pattern can be observed from our racial bias analysis. We trained the models on datasets skewed to be 70%, 80%, and 90% White. When tested on the unbiased test set, the accuracy patterns for each model remained identical: **Polynomial** models performed the best, followed by **Linear Regression** and then **KNN Regression**. Specifically, when tested on the unbiased test set, the **Polynomial Regression**'s error (RMSE) was consistently $\approx 9,066$, the **Linear Regression**'s error was $\approx 21,342$, and the **KNN Regression**'s error was $\approx 29,140$, regardless of which biased or unbiased dataset they were trained on.
- **Gender Bias:** Similarly, for the gender analysis, we trained one set of models on the original data and used them to predict on test sets with 70/30, 80/20, and 90/10 Male/Female splits. The mean predicted billing amount remained almost constant (e.g., $\approx 104,880$ for Linear Regression, $\approx 105,038$ for KNN), showing no significant change in the average prediction despite the skewed test data.

3. Algorithmic Bias

Finally, we analyzed algorithmic bias by calculating the error, in terms of RMSE for each demographic group using both biased and 90-10 split test sets. This allowed us to see if models produced more accurate predictions for some groups than for others

Contrary to the idea that a severely skewed dataset would lead to worse performance for underrepresented groups, the racial bias we introduced into the training data did not result in propagated bias. We found that the prediction error (RMSE) for each specific racial group remained constant, regardless of the training data's bias level. As shown in the below figure(fig. 1), the Polynomial model's error for White patients remained around 10,742, and for African American patients was $\approx 4,397$, even when trained on the 90% White dataset.

Figure 1: Group-specific RMSE on the unbiased test set for models trained on datasets with varying racial bias. Note that for each model (e.g., PolynomialRegression_D2), the error for each group (e.g., 'White') is identical across all four trained models, from 'unbiased' to 'biased_90'.



Yet, while propagated bias was not observed in our experiment, our model does show significant inherent bias across different demographic groups. We further concluded that the model's flexibility was a key factor in the severity of these performance disparities.

For Polynomial Regression, despite being the most accurate model overall, it was not necessarily the best model in terms of fairness as it produced significantly disparate error rates for different demographic groups. For example, its error rate for White patients (RMSE $\approx 10,742$) was nearly 2.5 times higher than its prediction error for African American patients (RMSE $\approx 4,397$).

For KNN Regression, it showed the worst performance disparities as it was extremely inaccurate for groups with fewer samples (Multiracial and Asian). With error rates of $\approx 50,000$ for multiracial patients, which is about 2.5 times larger than the model's best-performing groups.

Linear Regression, on the other hand, also showed notable performance disparities, with error rates ranging from $\approx 18,900$ for African American patients to $\approx 27,000$ for Multiracial patients. However, it demonstrated the smallest relative variability between its best and worst performing demographic groups, compared to much larger relative gaps in other two algorithms.

In conclusion, our experiments show that while the aggregated accuracy of our models was robust to our severely biased training data, the model themselves remains inherently biased.

This finding directly addresses our research questions as it showcases how introduced data collection biases (in terms of Age, Race, and Gender) did not propagate into our predictions nor did it harm the aggregate accuracy of the models. Our findings also suggest that the choice of model and its flexibility was the key driver of algorithmic bias. As shown in our results, the highly flexible KNN model produces some of the worst performance disparities, while the less flexible Linear Regression model had the smallest variability between its best and worst performing groups. Our experiments further conclude that while the Polynomial Regression model, which achieved the best performance in terms of accuracy, was not the fairest one.

Discussion

Findings

Our analysis aims to answer how data collection biases (Race, Gender, and Age) and model flexibility affect algorithmic bias in healthcare billing amount predictions. Our analysis led to two primary conclusions.

First, our primary finding was that data collection bias had a negligible effect on both aggregated accuracy and group-specific fairness. Contrary to the intuition that extremely skewed biased training data would lead to bias predictions, we did not observe propagated bias in our results. This finding suggests that the models did not learn to perform worse for the underrepresented groups. Moreover, as shown in Figure 1, the prediction error, in terms of RMSE, for each demographic group remained constant even for models that were trained on extremely skewed, biased datasets (i.e., 90% white). This further concludes that demographic under-representation was not enough to degrade our selected models' predictive performance.

Second, we found that while propagated bias was not observed, all three models exhibited significant inherent bias in terms of performance disparities. And the choice of model and its flexibility was the primary factor behind this gap. Specifically, the highly flexible KNN models that perform the worst in terms of accuracy also produced the most extreme and disparate error rates across different demographic groups. While the moderately flexible Polynomial Regression achieved the best performance in terms of accuracy, it also exhibits high performance disparities. The least flexible Linear Regression model, despite being less accurate, produced the smallest relative error gap between its best and worst performing groups.

These findings suggest how optimizing for aggregate performance metrics can sometimes obscure significant inherent biases in a model's performance. A model that performs best in overall accuracy may not be the most appropriate choice when fairness and equitable outcomes are our primary concerns.

Connection to Literature

- The [Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification](#) (Buolamwini & Gebru, 2018) paper found that facial recognition algorithms usually exhibit catastrophically higher error rates on dark-skinned women due to representational disparities. However, our findings show that under-sampling does not cause our models to exhibit higher error rates on minority groups.

Although the model and dataset complexity of our experiments are not comparable to those used by Buolamwini & Gebru, our findings still show that the harm of under-sampling is most severe when combined with highly flexible models.

Limitations and Future Work

While this analysis was conducted to the best of our ability, there is always room for improvement and areas that require further evaluation. The greatest limitation of all was the fact that the data we worked with was synthetically generated. We took many steps to ensure that our dataset was still viable for exploring bias, such as creating demographic distributions based on real-world data, and assigning weights to our categorical variables to create correlation based on research as to how they should affect our outcome variable, but there is no way to perfectly recreate the precision of actual data, and this must be kept in mind when reviewing our results.

Additionally, while the models we utilized had many advantages, they also have their own limitations and weaknesses. Linear regression is great because it is simple, but it also has a tendency to oversimplify complex relationships and assume additivity/linearity. The KNN model is very sensitive to data imbalance, and has the potential to overfit majority clusters or distort minority predictions. Polynomial regression is adept at handling complex relationships, but risks overfitting, and has a high sensitivity to outliers. If we were to return to this project in the future, it would be beneficial to compare more models in addition to these three, as a way to offset some of these limitations. For example we could incorporate random forests, as they have very low overfitting, as well as neural networks, due to their extensive ability to detect all possible interactions between variables.

One big issue with our methods relates to the way in which we split the data. We selected a 70/80/90% split to examine how our models would perform on extremely biased datasets, and while this was beneficial, we could have learned even more by incorporating a wider range of comparisons, both for extremes such as 99/1% splits or 100/0%, but also for reversing our splits. For example, we could observe the difference in model performance for 80% female participants vs. 80% male participants.

While our approach successfully demonstrates how model predictions can shift under changes in demographic composition, by incorporating a wider range of models, as well as a greater variation in the way we split our biased data, we would be able to significantly strengthen our findings and provide even more definite results.

Ethical and Societal Implications

Given our dataset was synthetically generated there were no ethical violations in the data collection process itself such as concerns regarding the data collection process violating patient privacy. However, because we introduced correlation into the data we have potentially introduced the risk of negative societal implications. For instance, to introduce correlation we conducted surface level research (i.e browsing the internet) to find trends between our features and outcome. Without conducting proper experiments or citing published research regarding the correlation between features and outcomes we are potentially mis-representing how different traits correlate with billing amount. Therefore, our work may suggest certain groups are being wrongfully discriminated against by the healthcare industry in turn making them less willing to seek out proper care.

However, because we are able to manipulate our data to discover correlations it allows us to also see the impact different biases have on predictive accuracy. We are able to replicate potential real-world biases without needing to access real patient's information and critique the harms that come with different biases. Additionally, because we are the ones manipulating the data the issue of accountability is designated to us or any future users. We know what correlations or biases are injected into the data, therefore the results of the algorithm and any effects-good or bad- are also designated to us.

Our work serves as a tool for validating hypotheses' and a warning to the harms of using biased data to train algorithms.